The Evolution of Conformity, Malleability, and Influence in Simulated Online Agents

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Abstract The prevalence of artificial intelligence (AI) tools that filter the information given to internet users, such as recommender systems and diverse *personalizers*, may be creating troubling long-term side effects to the obvious short-term conveniences. Many worry that these automated influencers can subtly and unwittingly nudge individuals toward conformity, thereby (somewhat paradoxically) restricting the choices of each agent and/or the population as a whole. In its various guises, this problem has labels such as *filter bubble*, *echo chamber*, and *personalization polarization*. One key danger of diversity reduction is that it plays into the hands of a cadre of self-interested online actors who can leverage conformity to more easily predict and then control users' sentiments and behaviors, often in the direction of increased conformity and even greater ease of control. This emerging positive feedback loop and the compliance that fuels it are the focal points of this article, which presents several simple, abstract, agent-based models of both peer-to-peer and AI-to-user influence. One of these AI systems functions as a collaborative filter, whereas the other represents an actor the influential power of which derives directly from its ability to predict user behavior. Many versions of the model, with assorted parameter settings, display emergent polarization or universal convergence, but collaborative filtering exerts a weaker homogenizing force than expected. In addition, the combination of basic agents and a self-interested AI predictor yields an emergent positive feedback that can drive the agent population to complete conformity.

Keywords
Filter bubble, recommender system, emergent conformity, opinion dynamics, polarization

1 Introduction

Humans have been subjected to hints, nudges and recommendations for centuries, but the widescale monetization of influence peddling began with the advent of paid newspaper and poster advertisement in the 1800s. That daily or weekly exposure to various merchants pales in comparison to today's continuous barrage of enticements and suggestions, some well-intentioned, others innocuous, and many strictly and cynically driven by the profit motives of both large corporations and the media platforms that sell them ad space (Wu, 2016).

For the vast majority of that formidable salesforce, diversity is not a friend. Although a soda bottler may tout 10 different varieties of its product (in an attempt to attract every conceivable consumer of carbonated sugar water), their job becomes much easier when everyone prefers one version, e.g., the *classic* recipe. Similarly, a media platform can pitch its ad spaces more convincingly

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with claims that the viewers (or listeners) in that space have a great deal in common and can thus be swayed with a single, well-placed promotion. Micro-targeting, though highly efficient in today's online world, is still a lot more resource demanding than simple blanket appeals guaranteed to reach and affect a homogenous audience. Diversity entails more work for sellers, while conformity promises much easier money.

Of course, consumer conformity also fuels fierce competition between sellers, all of whom must reach the same large group with similar products (though typically packaged and promoted differently). Diversity, on the other hand, carves out many niches and thus many different opportunities for sellers, who may slide into peaceful coexistence with one another. But for the ambitious, power-hungry actors that populate today's corporate landscape, little love is lost on competitors; their visions involve throngs of like-minded patrons sporting their clothing and consuming their beverages.

Winning this battle for popularity requires a mixture of reactive and proactive behavior. Companies must always respond to current trends, but anticipating future consumer attitudes and preferences, and positioning oneself accordingly, offers clear advantages in competitive markets. Hence, predictive skills are highly regarded and rewarded. Sports shoe producers who foresee a *green tide* and the popularity of recycled materials can begin retooling their production lines and rethinking their supply chains in advance of any seismic market shifts, and thus be ready and waiting when the first waves of environmentally-conscious athletes come ashore.

Of equal or greater significance is the ability of prediction to help *shape* future trends commensurate with the influencer's goals. Upon detecting an upswing in environmental awareness and predicting its consequences for their markets, a shoe manufacturer may get out in front of the wave by promoting the ecological policies of its main supplier countries, thus priming its present and future customers to favor their upcoming green product lines.

Prediction need not entail a strict temporal sequence. It also involves estimates of unknown information based on known facts. For example, an online retailer may predict an individual's clothing preferences based on their typical weekend locations: coffee shop, sports stadium or mountain trail. This, of course, may then aid in estimating their future purchases, but it may also assist in identifying them in a crowd photo taken days earlier. Both of these speculative tasks can evoke the label of *prediction*. The outputs of machine learning systems are often called *predictions*, even when they do not involve time.

Regardless of the exact temporal semantics of prediction, the influential powers that stem from anticipatory skill can also help herd consumers into a small number of bins, each well supplied with the influencer's products. Prediction aids influence, which can then be tailored to promote conformity, if so desired, as it so often is.

The (often unsolicited) tips, prods and suggestions that constitute so much of our daily lives generally reflect an ulterior motive of the *helpful* provider (of services or products). However, their end goal may not be a homogeneous flock of sheep, but merely a stable source of demand for their supply. Conformity emerges as a fortuitous by-product of influence, particularly of the well-researched, tailored and targeted variety. And as diversity dissipates, so too do the demands of research, tailoring and targeting. One size (of ad and product) fits all.

One suspected linchpin of this diversity reduction is data-driven similarity matching designed to expose an individual to suggestions that mirror the predispositions of other individuals *like them*. After completing an electronic book, a reader often cannot even swipe forward to the bibilography and appendices without first triggering a recommendation of other books that other readers with similar interests (revealed by a wide swath of their online behavior) have bought. This is known as *collaborative filtering*, and it seems fully capable of pushing a user population toward homogeneity. However, prominent researchers such as Massachusetts Institute of Technology lab co-founder Nicholas Negroponte (1996) argue that these systems expand the user's horizons and are key tools for the future of computing, and more recent publications by communication researchers (Borgesius et al., 2016; Bruns, 2019) claim that filter bubbles and echo chambers are poorly defined concepts that have no clear correlation with personalization and get far too much undeserved media attention.

Finally, comparative reviews of both simulations and empirical studies of personalization's effects reveal tendencies toward both increased and decreased diversity, depending upon a variety of local, regional and global factors (Keijzer & Mäs, 2022).

Alternatively, *content-based filters* examine the topics and authors previously chosen by a user and then suggest more of the same. At least on the surface, this appears to sort users into bins and then feed them the prototypical options within, thus restricting the diversity of each individual, though not necessarily damaging population-wide diversity: a multitude of bins may exist. Regardless of the filtering algorithm, and although most readers probably consider book recommendations useful, harmless, or, in the worst case, only slightly annoying, they may nudge readers toward conformity, one helpful hint at a time.

More grievous examples involve recommendations with well-disguised motives. Imagine a global-positioning system (GPS) for your car that, when asked to find the best route from A to B, favors the longer, slower route Y over the shorter, faster route X because (a) Y passes by many more fast-food restaurants than X, many of which have *bought influence* from the GPS provider, (b) your GPS history shows that you frequent such restaurants, (c) the time of day falls into your normal *hunger zone* indicated by your previous restaurant visits, and (d) GPS users with similar histories as yours have often patronized the establishments along Y. This is, unfortunately, not an entirely fictional scenario, as indicated by the general branch of advertising known as *geotargeting* (Wikipedia, 2022) and more specifically by the use of targeted applications (not simply passive ads) in the nascent field of automotive telematics, as decribed in Shashona Zuboff's (2019) influential book, *Surveillance Capitalism*:

> These are not ads on a screen but real-life experiences shaped by the same capabilities as targeted ads and designed to lure you into real places for the sake of others' profit. That means selling driver data to third parties that will figure out where you are, where you're going, and what you want. (p. 217)

Whether exposed or implicit, altruistic or devious, these influences fuel a powerful positive feedback cycle, which Jacob Ward (2022) has labeled *the Loop*, in which systems read and interpret our behavioral patterns and then use that information to steer our attitudes and actions. This circuit arises as individuals give in to the constant tug of influence, go with the flow, and thereby strengthen it, making resistance for themselves and others all the more difficult, while simultaneously enabling external actors to effortlessly maintain and enhance this cycle as prediction of individual behavior continually becomes easier. Excessive influence breeds conformity, which facilitates prediction, which simplifies control via further influence.

This article presents a very abstract model of social influence in an attempt to explore the emergence of global conformity from the local forces of peer-to-peer persuasion, alone and in combination with systems of wider scope: recommenders and predictors. The central focus is on the undesirable positive feedbacks that result between emergent homogeneity and ease of predictability: the cycle that increases human vulnerability to inducements.

2 Filter Bubbles and the Loop

Some of the earliest recognition of the disturbing effects of online forces upon human cognition and behavior came from Nicholas Carr (2010, 2014) and Eli Pariser (2011). Carr (2010, 2014) warned of the shallowness of thought and intellect that can result from an over-dependence upon technological aids to decision making, while Pariser coined the term *filter bubble* to denote the local information capsules that are gradually built around individuals as a result of ubiquitous personalizations crafted for us by online tools, often for purely innocent reasons, but with unfortunate consequences such as online *echo chambers*, in which conformity and loyalty are prized and dissent punished.

Pariser (2011) characterizes internet filters in rather disturbing terms:

> They are prediction engines, constantly creating and refining a theory of who you are and what you'll do and want next. Together, these engines create a unique universe of information for each of us – what I've come to call a filter bubble – which fundamentally alters the way we encounter ideas and information. (p. 9)

Both Carr (2010, 2014) and Pariser (2011) fear significant declines in individual creativity, open-mindedness and general concentration and learning abilities caused by the internet of influencers, both human and virtual. But, as mentioned earlier, many experts remain unconvinced of the causes, severity or even existence of filter bubbles and echo chambers (Borgesius et al., 2016; Bruns, 2019; Keijzer & Mäs, 2022; Negroponte, 1996). This controversy only accentuates the need for further research into diversity reduction—both its causes and consequences—and into feedback loops where consequences become causes, and vice versa.

In a recent book, *The Loop*, Ward (2022) discusses positive feedback cycles whereby conformity breeds greater susceptibility to online influence, which pushes groups further toward homogeneity. Ward describes an interlocking system of loops involving three main entities: humans, corporations and artificial intelligences (AIs). Our species has a portfolio of fundamental behaviors passed down through many branches of the evolutionary tree, and once our difficulties in overriding some of these innate mechanisms became known, merchants began exploiting them. They do so by observing people and then feeding influential stimuli (expertly designed for maximal effect) back into large segments of the population. A higher-level loop involves AI systems and their ability to recognize behavioral patterns that are too complex for most human observers to detect. In addition, these patterns can encompass both primitive, innate tendencies and more sophisticated, context-sensitive and idiosyncratic acts. The AI can thus parse sequences of both impulsive and reasoned activities, and learn what stimuli work best for **you**, not just your cohort.

Ward (2022) fears for the future of such a system, since it embodies less of a negative feedback cycle that could evolve to a healthy steady state and more a set of positive feedbacks that can eventually devolve to a very homogeneous, uncreative, predictable, and thus too-easily-controlled humanity:

> The Loop is a downward tailspin of shrinking choices, supercharged by capitalistic efficiency, in which human agency is under threat from irresistible systems packaged for our unconscious acceptance....we're guided by unconscious tendencies, but we rarely detect it when they are analyzed and played upon. Now throw pattern-recognition technologies and decision-guidance strategies at us. And do it all in a society that doesn't have the long-term sensibilities in our policies and in our programming to recognize and regulate something that will determine the future of our species. (p. 17)

Surveillance Capitalism (Zuboff, 2019) also addresses this imperceptible push toward conformity and controllability:

> ...the worker will adhere to norms, the group will swarm to defeat anomalies. We will all be safe as each organism hums in harmony with every other organism, less a society than a population that ebbs and flows in perfect frictionless confluence, shaped by the means of behavioral modification that elude our awareness and thus can neither be mourned nor resisted. (p. 410)

The shared, and central, drawback of Ward's (2022) loop and Pariser's (2011) filter bubble is a narrowing of options for both individuals and the population as a whole. Although most modeling efforts in this arena have focused on the filter bubble and its detailed structure and/or the process by which it emerges, this article investigates Ward's (2022) loop as well. In many studies, whether real-life social or model-based synthetic, researchers monitor diversity (of each agent's interests/preferences and/or those of the population as whole) and how it changes under the influence of various forms of personalization and recommendation. This article takes a step beyond to explore the emerging positive feedbacks of these systems and the role played by prediction: an influencer's ability to ascertain an agent's behaviors.

3 Related Work

3.1 Overview of Opinion Dynamics

The study of opinion dynamics and the emergence of homogeneity, polarization and diversity has a long history in the social sciences. Several articles provide insightful overviews and perspectives (Axelrod, 1997; Centola et al., 2007; Feliciani et al., 2017; Flache et al., 2017; Giardini et al., 2015; Keijzer & Mäs, 2022). This section gives a vertical cross-section of empirical and modeling studies of greatest relevance to the models developed in subsequent chapters.

A key underlying factor in most of this research is *homophily*: the tendency of people to associate primarily with and be influenced by individuals with whom they already have a lot in common (McPherson et al., 2001). Online personalization magnifies homophily by exposing us to the preferences of very large groups of similar individuals, much larger than one's normal network of friends, relatives and colleagues. Whether the matchmaking wizard is Google, Amazon or Facebook, the same concerns over ideological convergence arise (Lazer, 2015), despite the lack of concrete empirical evidence (Keijzer & Mäs, 2022) that homophily leads to widespread conformity.

Mathematical modeling of homophily dates back to the 1950s (French, 1956), while the advent of the internet has spurred considerable interest in the topic from the mid-1990s to the present. The history of these models has been thoroughly reviewed by Flache et al. (2017). They typically involve a collection of *opinion* variables for each simulated agent, with some of the key differences between the models including:

- the use of a single or multiple opinion variables per agent,
- the use of nominal or metric values for each opinion,[1]
- the potential for positive (assimilative) influences, negative (repulsive) influences, or neither (neutrality) as the result of an agent–agent interaction,
- the use of *bounded confidence* to influence, wherein agents with an opinion difference (using various metrics) above this bound cannot affect one another,
- the presence of interaction noise, which occasionally allows highly dissimilar agents to interact and sway opinions,
- the static or dynamic nature of the interaction network among the agents, and
- the explicit modeling of message passing or broadcasting.

DeGroot's (1974) model is perhaps the simplest. It uses the weighted opinion average of agent A's neighbors to compute A's own opinion; and through repeated updates, the population converges on a shared opinion. The weights attributed to neighbors are fixed and not directly influenced by their opinions, so the model lacks homophily, which requires influential power to correlate with agent similarity. Despite this simplicity, the model's core mechanism of weighted opinion averaging has spawned many variants.

1 Nominal values are discrete and have no underlying distance metric, so nonequal values have no measure of partial equality or nearness. Conversely, metric values can be discrete or continuous and have a well-defined distance formula for assessing equality and degrees of similarity.

One of the most influential models comes from Stanford's Robert Axelrod (1997), who wondered how societies could ever maintain ideological diversity in the face of continual interaction and its homogenizing influences. Indeed, the early models (Abelson, 1964; DeGroot, 1974; French, 1956) showed strong tendencies toward full convergence in association networks with connectivity patterns in which every pair of agents (A and B) has a communication chain from A to B, B to A. or both. However, those models use only a single trait to characterize each agent, whereas Axelrod (1997) employed a full opinion vector of nominal values and a two-dimensional lattice (grid) on which agents reside and interact, but only with their neighbors.

Axelrod's (1997) model incorporates a confidence modulator (as opposed to a strict threshold) to realize homophily: the probability that a neighbor (N1) will affect another (N2)—by changing one of the N2's opinion values to that of N1—is proportional to the number of matching opinions between N1 and N2. Influence spreads gradually throughout the graph, but in less-than-intuitive ways. One of Axelrod's key findings was that complete homogeneity was most likely to emerge in both small and large lattices, but not in those of intermediate size, e.g. 10×10 cells. This interesting discrepancy stems from the complexity of interactions among multivariate opinion vectors, which, of course, are more realistic than single-valued characterizations.

However, as discovered later by Flache and Macey (2006), another key factor to diversity maintenance in medium-sized Axelrod scenarios was the *nominal* nature of opinion variables; even though each vector component can take on several values, no ordering or distance measure (a.k.a. metric) exists among them. Hence, any two values either match perfectly or conflict, with no partial similarities. Neighborhood deadlocks then occur when agents are in complete disagreement with their neighbors, and hence cannot be influenced by them. When Flache and Macey used metric (but still discrete) value spaces, the ensuing partial similarities avoided most deadlocks, thus allowing homogeneity to emerge in almost all simulations. Their only constraint on opinion convergence was the confidence threshold, which could be set high enough to greatly curtail influential interactions and thereby preserve diversity. Later work (De Sanctis & Galla, 2009), illustrated emergent homogeneity across a wide range of confidence bounds while also indicating the homogenizing effects of small and intermediate levels of interaction noise.

The importance of *bounded confidence* to consensus formation was recognized by several researchers in the late 1990s and early 2000s. Various models are nicely summarized, categorized and tested by Hegselmann and Krause (2002). They frame the potential for influence as weights on the connections between agents, with higher weights reflecting greater agreement of opinion between the two. A confidence bound (ϵ) represents a threshold on the opinion difference between two agents. If that difference exceeds ϵ, the weight on the corresponding agent–agent link essentially becomes zero, and thus, the two cannot interact. Since opinion vectors change, so too do the paired similarity values, and thus the weights. The whole topology of the connection network thereby becomes dynamic in that the functioning connections between neighbors change over time. The various settings of confidence bounds can then create different emergent outcomes, from consensus (i.e., complete homogeneity) for large ϵ, to a few well-differentiated opinion clusters (i.e., polarization) for intermediate ϵ, to high fragmentation (i.e., diversity) for small ϵ.

Another important local factor is the ability of vastly different agents to sway opinions in a direction opposite to their own, i.e., to *repel*. Formalized by Rosenbaum (1986), the *repulsion hypothesis* argues that emerging polarization stems more from the repulsion of disagreeing agents than from the convergence of like-minded individuals to a few well-spaced cliques. A pair of psychological studies involving physical human attributes and political affiliation, respectively, showed that subjects (college students) were much more strongly repulsed by others whom they found unattractive or ideologically opposing than they were drawn to the attitudes of those whom they found attractive or agreeable. Based on these and many other cited studies, Rosenbaum posited that encounters between similar individuals provide little motivation for change, whereas those among disparate agents invoke more arousal and stronger feelings, which often motivate repelling change.

Conversely, Mäs and Flache (2013) provide simulations in which polarization emerges solely from the forces of attraction; no local repulsion occurs. They review a host of empirical and modeling studies and observe that in situations where (a) opinions have continuous values (e.g., degrees of agreement with a particular issue) and (b) polarization emerges, there is often a repellent possibility in the agent's interactions. Their model achieves similar emergence, but without repulsion. This is a noteworthy result, since continuous-valued (and thus metric) opinion spaces avoid the deadlocks associated with nominal values; and without these inter-agent opinion roadblocks, a freer transmission of views often gradually leads to full consensus, not polarity.

Mäs and Flache's (2013) model relies upon the caching of each agent's history of (received) influences; their weighted average determines the agent's opinion; it has no other predilection. This full dependence upon recent arguments, with no anchoring in one's previous views, creates a strong enough force to separate populations into distant camps, even those that begin as nearly homogeneous mega-clusters.

Work by Centola et al. (2007) shows a similar trend toward diversity, despite the absence of inter-agent repulsion. However, their model leverages both nominal opinion spaces (which provide considerable protection against homogeneity) and dynamic communication networks (which allow agents to disconnect from conflicting neighbors and find others more to their liking). Their model is essentially Axelrod's (1997), but enhanced with dynamic neighborhoods. This empowers agents to construct their own echo chambers, which naturally leads to polarization, even though no agent's opinions receive negative influence from disagreeable encounters.

Somewhat at odds with the preceding findings (Centola et al., 2007; Mäs & Flache, 2013), the graph theoretic proofs of Dandekar et al. (2013) indicate that emergent opinion diversity requires more than basic homophily. They attempt to add more social realism to the interaction model by assuming that individuals are strongly biased toward their original positions and will therefore more readily accept confirming than refuting evidence, though they can accept either. They supplement DeGroot's (1974) original model of weighted neighbor opinion averaging with a measure of *biased assimilation*. DeGroot's model, with or without homophily, is shown to converge to full consensus, while the addition of biased assimilation promotes polarization.

The models discussed so far involve opinion feedback, but solely via local agent-to-agent interaction. The emergence of cultural cliques in the form of agent groups with very similar opinion vectors then have interesting collective force, but no explicit global mechanisms exist in these systems. However, the real world consists of many actors capable of both gathering information from many sources and disseminating it to myriad targets. Chief among these is the media, whether conventional or social. Intuitively, it would seem that any entity with mass global influence could quickly promote homogeneity.

Surprisingly, one of the earliest variations of Axelrod's (1997) model (Shibanai et al., 2001) revealed the opposite effect. Their system computes a running average of opinion values across the entire agent population and essentially employs this global average agent (GAA) as an extra neighbor to every standard agent. When chosen for an opinion update, each agent then stochastically choses a neighbor from which to receive confidence-modulated influence. The odds of choosing GAA as that neighbor is a key parameter of the system, λ. Runs of a 10×10 Axelrod model showed that larger values of λ yielded faster convergence (as expected), but to higher levels of diversity (counterintuitively) than runs with lower λ settings. Apparently, high pressure toward convergence (to the population mean) quickly induces a cultural clique, but one containing only a subset of the entire population. That clique then tends to dominate the global average, which feeds back to (a) push the clique members even closer to its mean, but (b) without significantly influencing non-clique members, since their limited overlap with the mean yields low confidence values and thus few interactions with GAA and clique members. So as the clique members converge to a mean, they also diverge from the non-clique members. Their results also show that rapidly-forming cliques are significantly smaller than those congealing more gradually. When $\lambda = 0$, global homogeneity arises, albeit very slowly.

Keijzer and Mäs (2022) discuss the *personalization-polarization hypothesis*, which involves a different entity with global outreach: a recommender system that can gather considerable data from many agents and then average and generalize from it to produced targeted suggestions for other agents.

They address the concern that these idiosyncratic tips promote polarization, but they find anecdotal, empirical and modeling evidence to both support and refute the hypothesis. They call for further, and more thorough study, with agent-based, complex-systems simulations as a primary tool. To wit, the emergent patterns in complex social networks can hinge upon intricate local behaviors, some of which cannot be abstracted away nor aggregated into population-level variables. The agent-level details really matter. This has made the social sciences one of the more popular venues for complex-systems research.

In the same article, Keijzer and Mäs's (2013) simulations illustrate the diverse effects of communication protocols upon polarization. When global entities can broadcast the same message to many agents at once, a deep opinion chasm can easily form; whereas simple agent-to-agent messaging lacks this polarizing power, since any agent that is pushed in an extreme direction by a single encounter can quickly return to the status quo after subsequent interactions with its other neighbors. With broadcasting, many of those neighbors can be pushed to extremes simultaneously, thus destroying the stabilizing base that helps rein in outliers.

In Cinelli et al. (2021), an empirical analysis of over 100 million posts and interactions on several of the most popular social-media platforms reveals a clear connection between (a) global information handling and targeted dissemination, and (b) echo-chamber formation and polarization. The authors' comparison of Facebook and Twitter (which have centrally-controlled newsfeeds), with Reddit and Gab (which allow users to adjust their feeds) reveals much more pronounced echo chambers on the former than the latter, although a viable counterargument is that the latter are already liberal (Reddit) and conservative (Gab) clusters with few additional opinion differentials that could widen to full-scale divisions.

Dandekar et al.'s (2013) work on biased assimilation (discussed earlier) also includes a model of recommender systems and their influence. Using three simple, elegant, random-walk algorithms on bipartite graphs, they show that three common recommender algorithms can all lead to polarization under realistic parameter settings.

Fleder and Hosanagar (2009) also employ a sophisticated model (similar to Dandekar et al., 2013) with random-walk dynamics. They simulate eight different product-suggestion algorithms (most of which are collaborative) to explore a pair of seemingly conflicting opinions about collaborative filters: (a) they point out new possibilities to consumers and thus broaden their horizons, increasing diversity, versus (b) they merely increase the exposure of already-popular alternatives, thus embodying a positive feedback that decreases population-level diversity. The authors characterize diversity via the Gini coefficient (Gini, 1921) of the distribution of products sold by a company. The evolution of the Gini ratio across a long series of consumer choices then indicates diversity changes. By comparing scenarios with and without recommender-based influences, the authors show that for the collaborative filters, the nudges do indeed produce a positive feedback, but one that eventually reconciles the two opposing hypotheses about collaborative filters: when suggestions are employed, diversity of individual purchasing portfolios increases, but population-wide diversity declines. In short, users become more internally complex while simultaneously becoming more similar to others. Conversely, several of their recommender algorithms behave non-collaboratively, such as by promoting the least-popular items, and these lead to global diversity, as expected.

Nguyen et al. (2014) expanded the scope of Fleder and Hosanagar's (2009) work by considering not only how the users' choice distributions change over time, but also how the distribution of recommendations evolves. Their work examines a large corpus of movie recommendations and choices from MovieLens,[2] which has hundreds of thousands of users, produces millions of ratings, and employs collaborative filtering to make suggestions. The authors had access to both the movie ratings given by users and the recommendations that they received. When a user rates a movie recommended to them, the system can infer that the user took the recommendation. This allowed the authors to partition users into a *follower group* that takes recommendations, and an *ignorer* group that does not.

2 https://www.movielens.org/.

Furthermore, users judge the appropriateness of a large set of features for each movie, yielding comprehensive *tag genomes* for each movie. Distances in the high-dimensional tag space can then quantify the similarity between movies; and the pairwise average similarity among a set of movies serves as their diversity metric, which can then be applied to both sets of recommendations and sets of user ratings, and for both the follower and ignorer group. This elegant framework supports a straightforward assessment of the influence of recommendations upon diversity. To wit, diversity decreased over time for both the follower and ignorer groups, with the latter experiencing a more significant decline. Thus, following recommendations seems to reduce the power of the filter bubble, though it still forms. Furthermore, the movies recommended to both groups had sinking diversity levels, with a less dramatic narrowing effect for follower than ignorer recommendations. In short, this work indicates a tendency (in a large corpus of real-world data) toward decreased diversity in both user preferences and the recommendation sets provided to them, but collaborative filtering may actually mitigate the homogenization process to some degree.

Gottron and Schwagereit (2016) perform a detailed, large-scale, agent-based simulation of filter-bubble formation on a dynamic communication network. Their runs begin with an abstract social network (among user nodes) whose connection topology grows via *preferential attachment* (Barabasi, 2002). The user nodes then send messages to their network neighbors with varying content determined by the topic-preference vectors of the sender. Receiving agents then rate the relevance of incoming messages based on their own topic vectors. Monitoring each user is a personalization algorithm, which tries to learn its user's topic preferences. This algorithm acts as a filter on incoming messages, releasing only the best K of them to the user, with its ranking based on the personalizer's understanding of the user's preferences, a probabilistic model that it gradually refines by examining the (filtered) messages that the user accepts and rejects. Thus, a feedback loop between a user and its personalizer creates a dynamic situation that can gradually build a filter bubble around each user, which they define as a filter that becomes pickier about the messages it accepts in terms of the message's sender and/or topics.

The internal topic vectors of users remain static, but the authors explore different settings for the parameters that control how those vectors are used to determine relevance, essentially controlling how strictly the agent rejects (filtered) messages that poorly overlap with its high-priority topics. As expected, with high user strictness, the personalizer adapts to form a strong filter bubble. More interesting, but also quite intuitive, is the fact that well-connected users tend to engender stronger filter bubbles, since their filters have more messages to choose from and thus have few peripherally-relevant messages in the top K. In general, they find that filters keying off of author information tend to create stronger bubbles than those using topical content. However, the key contribution of this work is the model itself, which captures important interactions between user biases, generated and accepted messages, and adaptive personalized filters.

Min et al. (2019) take a more direct and active approach to filter-bubble analysis by creating many (128) social bots, each consuming a particular topic, and letting them loose on a blogging network. Based on messages received and preferred, the bots gradually modify the sources that they follow, thus changing the local topology of their social network, which, in turn, affects the messages that they receive. However, no recommenders or personalizers are involved; all filtering is user-driven by the followings that they create. After several months of operation, the authors examined the degree to which the information feeds to each agent became polarized and how the evolving network structure may have caused that narrowing of exposure.

They found less polarization in bots specialized for science and technology messages than for those focused on entertainment, despite the fact that the local topologies for science bots displayed considerably higher clustering (and thus more cliques), which would, intuitively, indicate a strong structural bias toward polarization. However, a more detailed comparison of the topologies for science versus entertainment bots revealed different macro structures in which messages from localized science clusters tended to be transferred to central nodes that broadcast them to many other clusters, thus combatting polarization; whereas entertainment networks had far fewer clusters and, most significantly, fewer inter-cluster pathways. In general, this research uncovers interesting relationships between dynamic network structures and filter-bubble genesis while using standard metrics for bubble strength and network characteristics (such as clustering, density and motifs).

More recently, Bagnoli et al. (2022) examined the effects of recommendation upon filter-bubble formation in a simulated social network. They represent agents as simple perceptrons whose inputs consist of several factors (present in messages) and which output a rating in [−1, 1]. The recommender has access to the perceptron weights of each agent and can thus compute all of their outputs, which it aggregates into an opinion matrix (**M**). On each timestep, a random user is selected to produce a message (*m*), whose contents derive stochastically from the weights of its perceptron. For each such m, the recommender uses **M** to determine which users will favor m; only they will then receive it. When the model runs in learning mode, each such agent then updates some of its weights to more properly align with the factor values in m, i.e., the agent's views on the relative importance of some factors are nudged in the direction that more closely mirrors the contents of *m*.

By varying parameters such as (a) the number of initial messages (generated by the system itself, not by the individual agents), (b) the threshold (τ) for determining whether to send m to a given agent, (c) the number of non-zero factors in messages, and (d) the learning rate, the authors show predictable causal relationships. For example, a higher learning rate leads to more pronounced polarization, as does a high τ value, even when no learning occurs. As in Gottron and Schwagereit (2016), this work presents a useful model that could be extended and integrated with other models in many ways, although the results so far are largely intuitive.

Finally, the PhD thesis of Mingkun Gao (2021) provides a lengthy overview of *attitude-consistent information consumption* or *selective exposure*, which ties directly to filter bubbles and echo chambers. The project's main goal is the design of social media interfaces that promote the sharing of diverse, often attitude-inconsistent, views in order to combat polarization and homogeneity. Gao also digs into the role that individuals play in creating (often inadvertently) filter bubbles for their online peers. The thesis provides an extensive summary of related research into the narrowing scope of information exposure (in its various guises) and the diverse attempts to ameliorate it.

3.2 Brief Comparison to the Current Models

The models introduced in this article have many similarities to those described above. They employ a very abstract representation of an agent and its opinions: a vector of real-valued, generic characteristics, i.e., *multiple metric traits*. Agent interactions are simple random pairings in which agent B may modify one characteristic of agent A if either A and B are similar (*assimilation*) or very dissimilar (*repulsion*), and a pair of parameterized *confidence bounds* serve as gated preconditions for both positive and negative influence. The model involving a recommender system uses *interaction noise* as occasional random interactions supplement the strategically chosen pairings of the personalizer. None of the models below use an explicit *interaction network*, although implicit biases among the possible pairings do arise in the recommender scenarios.

Unlike some of the more detailed models above, the agents in this work do not produce *messages*, so no global filter decides who gets what message. Only a local comparison between agent vectors governs whether and how one may influence the other.

Although the precise combination of model features listed above is possibly unique, the primary novelty of this work is support for a simple predictive task, thus enabling simulations of the feedback loop between population diversity and the predictive opportunities of online actors. This facilitates a more direct investigation of Ward's loop, with peripheral implications for filter-bubble research.

4 A Basic Model of Social Influence

The following model captures interactions between two individuals, A and B, of the same basic type, i.e., *peers*.[3] For each such pairing, at most one of the individuals will change. If A and B are

3 All code used for the simulations in this article is publicly available at https://github.com/keithlinndowning/conformer.

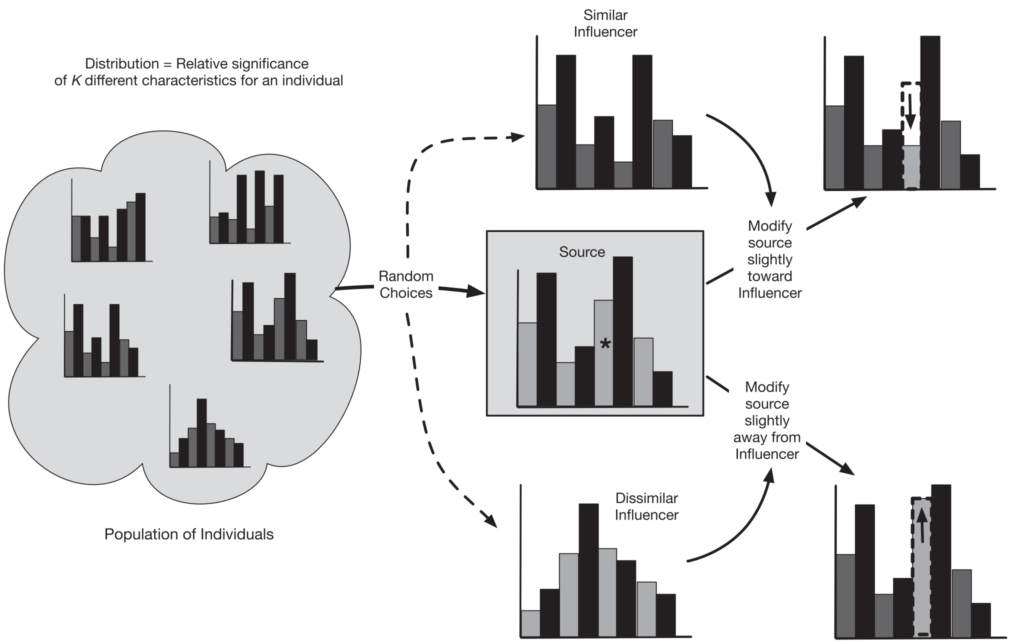

Figure 1. The basic model of influence involving a population of distributions that serve as the basis for composite characteristic vectors (CCVs). At each timestep, one source and one influencer are randomly drawn from the population. If their CCVs are very similar (upper path) or very different (lower path), a random characteristic (starred) of the source is chosen and modified toward (upper) or away from (lower) that same characteristic of the influencer.

similar, then certain aspects of A's attitudes and/or behaviors can be modified, ever so slightly, in the direction of B's. This captures a simple act of homophily, peer-to-peer *recommending* and acceptance of that suggestion. Conversely, when A and B are very different, B's recommendation may cause A to change, but in opposition to B's advice, thus exhibiting repulsion. For example, when an extreme liberal meets a staunch conservative, they may exchange pleasantries, but any literature or upcoming events mentioned by one is more likely to be explicitly avoided than pursued by the other.

As shown in Figure 1, the model represents each individual as a normalized distribution over a fixed set of continuous-valued characteristics. This will later become separate distributions for attributes and behaviors, but the basic algorithm is most easily illustrated with a single distribution. At each timestep, a *source* individual is randomly drawn from the population, as is a potential *influencer*. One of three local events then occurs:

1. Assimilation: If the source and influencer have very **similar** distributions, then one randomly-chosen characteristic of the source is altered to move **toward** the value of the influencer's corresponding characteristic.
2. Repulsion: If the source and influencer have very **different** distributions, then one randomly-chosen characteristic of the source is altered to move **away** from the value of the influencer's corresponding characteristic.
3. Neutrality: Lacking significant similarity and divergence between the two, the influencer has no effect upon the source.

Formally, each distribution contains only non-negative values and is continually normalized (after each modification) to a vector of length 1. The similarity of two individuals stems directly from the angle between their vectors, with angles within θ_1 of 0° denoting similarity, and those within θ_2 of 90° signifying divergence. For most of the runs described below, $0° \leq \theta_1 \leq 27°$ and $\theta_2 = 27°$. Smaller values of θ_2 give very infrequent matches, since the inter-vector angles rarely approach 90° in the models described below.

The composite characteristic vector (CCV) described above and depicted in Figure 1 is derived from two component unit vectors, representing attributes and behaviors, respectively. An individual computes its CCV as the geometric norm over the merged attribute and behavior distributions. Similarity assessments of two individuals employ the CCVs of each, but any modifications to the source (toward or away from the influencer) involve a single element of either the attribute or behavior vector. In an upcoming section, the significance of the attribute-behavior division becomes clear when the model is enhanced to include a *predictive* agent.

For any character c, the source's (s_c) and influencer's (i_c) values combine to update s_c as follows:

$$s_c = s_c + z\alpha(i_c - s_c) \tag{1}$$

where α is an adaptivity rate and z is either +1 or −1 depending upon whether the source and influencer are very similar or different, respectively. Unless otherwise stated, $\alpha = 0.25$ for the runs reported below.

At the start of each run, the attributes and behaviors of all individuals are initialized to random real values in [0,1]. Both vectors are then normalized geometrically and separately, and the concatenation of those normals is normalized again to produce the CCV. This process produces a population with an average inter-CCV angle of approximately 40°, with a standard deviation of approximately 8°. Thus, in the initial population, the vast majority of angles fall in the (2 *SD*) range $[24°, 56°]$.

Simulation of this basic model proceeds as follows:

1. Initialize a population of P individuals to random attribute and behavior vectors.
2. Repeat for T timesteps:
   - Select a random source (S) and influencer (I) from the population.
   - Compare the CCVs of S and I.
   - If S and I are very similar or different, modify one characteristic of S accordingly.
   - Compute and save population diversity
3. Plot the history of population diversity.

The diversity measure between two individuals is simply $1 - cos\gamma_{i,j}$, where $\gamma_{i,j}$ is the angle between the CCVs of individuals i and j. Population diversity (Δ_p) is thus the average of the diversities between each pair of individuals:

$$\Delta_p = \frac{\sum_{i \neq j}(1 - cos\gamma_{i,j})}{P(P-1)} \tag{2}$$

Given the initialization procedure described above, the average inter-CCV angle of 40° yields a starting diversity of approximately 0.23. Thus, the population begins in neither a maximally nor minimally diverse state, but rather, an intermediate state that one might expect from a population of agents who begin with initial propensities chosen both randomly and independently of one another and of those of other agents.

In summary, the basic model of social influence uses purely local, *peer interactions* between individuals as the basis for making asymmetric modifications (i.e., to only the source agent), thus nudging one agent slightly closer to or away from the other.

Table 1. Parameter settings for several runs of the basic influencer model.

Parameter	Value
Number of characteristics	15
Population size (P)	50
Total timesteps (T)	250,000
Adaptivity rate (α)	0.25
Similarity trigger proximity (θ_1)	$\{27^\circ, 36^\circ\}$
Difference trigger proximity (θ_2)	$\{18^\circ, 27^\circ\}$

Table 1 lists the settings of the main parameters for a series of runs of this simple model. The key differences between these runs are the values of the two trigger proximities for peer influence, θ_1 and θ_2. As seen in Figure 2 (upper left), when both are 27°, the population becomes more diverse, since peer influences both assimilate and repulse. However, reducing θ_2 to 18° induces conformity (Figure 2, bottom), as the positive (assimilative) peer influences dominate the negative (repulsive). This conformity involves a diversity value between 0.04 and 0.14, which corresponds to an average inter-vector angle between 16° and 31°. In these runs, convergence requires around 50,000 timesteps (i.e., 20% of the total runtime). This means that, on average, each of the 50 agents plays the role of source 1,000 times during the transient period.

In the 25 runs that produce significant convergence (Figure 2, bottom), the population forms 5–10 clusters in vector space, thus reflecting multi-modal polarity rather than perfect conformity. However, by increasing θ_1 to 36°, complete homogeneity (i.e., a single cluster housing all agents) does emerge, as revealed by the steady-state value of $\Delta_p = 0$ in Figure 2 (upper right). Looking again at the bottom of Figure 2, note that the low repulsion trigger ($\theta_2 = 18^\circ$) reduces negative influence, and, in fact, none occurs in any of the 25 trials. Thus, homophily has full rein to drive the population to homogeneity, but it fails to do so. This violates naive expectations but fully aligns with many of the studies cited earlier, wherein homophily produces clusters with little interaction and thus no merging.

An important dynamic underlies these runs of the basic model: the onset and development of both assimilative and repulsive peer influence. As discussed earlier, the initialization procedure for CCVs produces inter-CCV angles that primarily fall in the range $[24^\circ, 56^\circ]$. Thus, when $\theta_1 = \theta_2 = 27^\circ$, only a few initial pairings should be eligible for positive interactions, but even fewer will support negative influences. Hence, the population should initially have a slight tendency to cluster rather than disperse. Figure 3 reveals this trend, as the number of positive interactions per epoch rises dramatically in the first 250 epochs, while the degree of negative persuasion stays below 5 until around 500 epochs, when it slowly rises and eventually steers the population toward a diversity level higher than its initial value (of approximately 0.23). It seems that the initial (weak) tendency to pull CCVs toward one another also draws them apart from other vectors such that, eventually, the number of strongly conflicting CCV pairs invokes a powerful enough negative feedback to dissolve many of the clusters. As shown in a later section, this same general dynamic plays out when the basic model is supplemented with a goal-oriented, predictor agent.

In summary, runs of the basic model indicate that, on their own, local interactions between basic agents (a.k.a. peers) that occasionally influence one another can decrease diversity, to either polarization or complete homogeneity. However, this only occurs when the positive forces of peer influence outweigh the negative, or, in societal terms, when individuals are more likely to interact

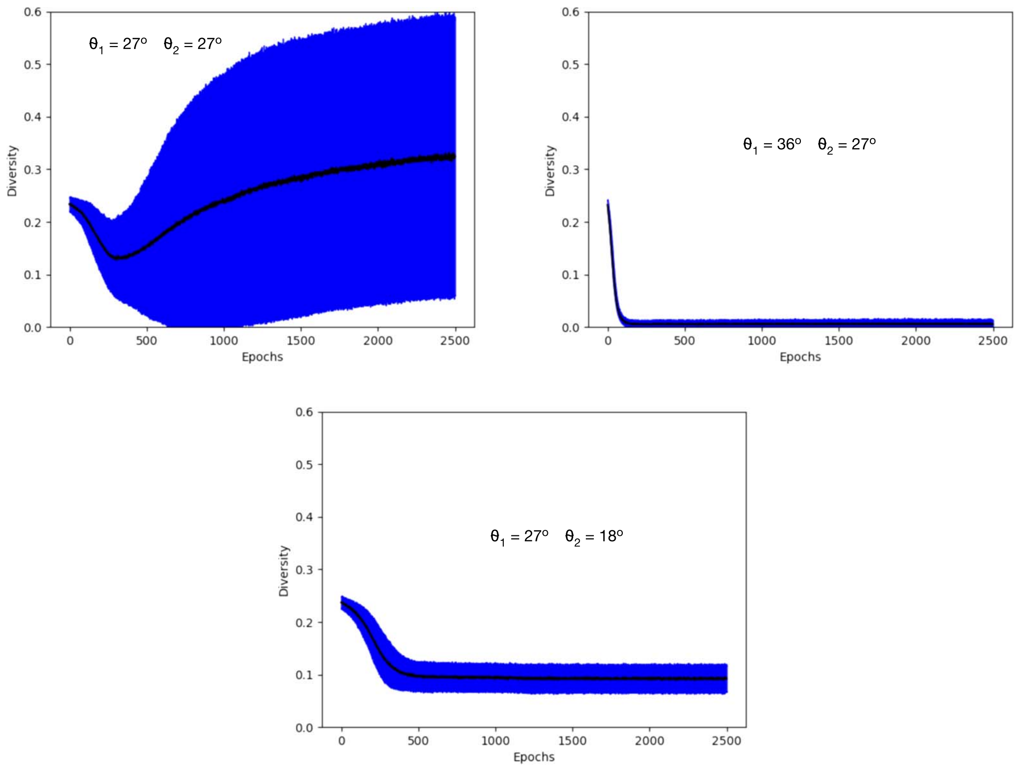

Figure 2. Time series of population diversity (Δ_p) averaged over 25 runs for three different scenarios of the basic model of social influence. All runs employ the parameters given in Table 1. (top left) $\theta_1 = \theta_2 = 27^\circ$ produces divergence. (bottom) $\theta_1 = 27^\circ, \theta_2 = 18^\circ$ yields multimodal polarity. (top right) $\theta_1 = 36^\circ, \theta_2 = 27^\circ$ creates perfect homogeneity. Each epoch consists of 100 timesteps, and shaded regions delineate standard deviations for the Δ_p values at each time point across the 25 runs.

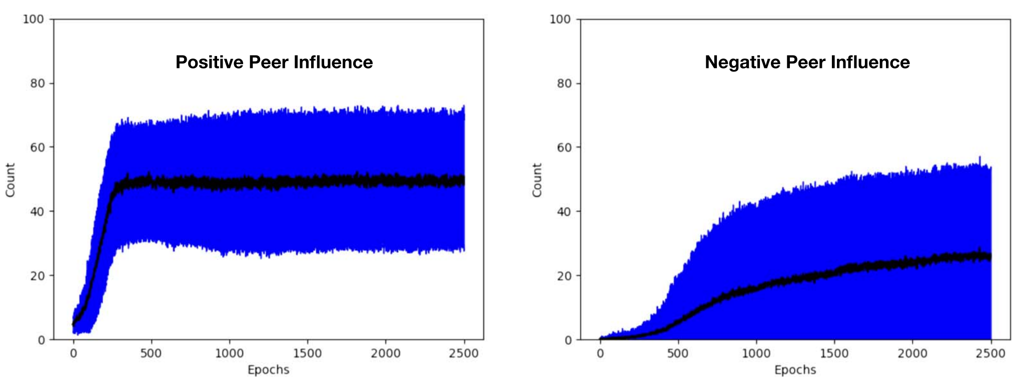

Figure 3. For the divergent (top left) diagram of Figure 2 (where $\theta_1 = \theta_2 = 27^\circ$), the fall, then rise of diversity correlates with the onset of (left) positive peer influence, followed later by that of (right) negative peer influence. Both plots show the averages and standard deviations over 25 runs of the number of instances of peer influence per epoch.

and be persuaded by similar agents than to encounter and react adversely to opposing viewpoints: when homophily dominates repulsion. Yet even when positive influences overwhelm the negative, there are no guarantees of complete homogeneity.

5 The Matchmaker Enhancement

Consider a second model that replaces all random local interactions with a global matchmaker akin to an online recommender system. On each timestep, the matchmaker randomly selects an individual from the population (as the source) and then **strategically** selects the influencer as that individual most similar (according to CCV comparisons) to the source. The source and influencer then interact as before, subject to the restrictions of θ_1 and θ_2, although the latter (repulsive) trigger rarely comes into play in this scenario. Using $\theta_1 = 36°$ and $\theta_2 = 27°$ provides a good comparison to the purely local, peer-based model, since those triggers yielded complete homogeneity, as displayed in Figure 2 (upper right).

Once again, the naive assumption is that the matchmaker will easily drive the population to homogeneity, especially (a) in such a simple model with so few agent attributes (compared to humans), and (b) in the same scenario that achieved complete conformity from only local interactions. But this is not the case. Instead, as shown in Figure 4 (upper left), it quickly (over 5,000 timesteps or less) yields stable diversity levels in the range [0.13, 0.21] (the average plus 2 *SD* in each direction). These diversities correspond to an inter-vector angle range of approximately $[29°, 37°]$. Hence, the matchmaker partitions the population into numerous, diverse cliques by nudging individuals in the direction of those most similar to themselves. Presumably, once agent A moves toward

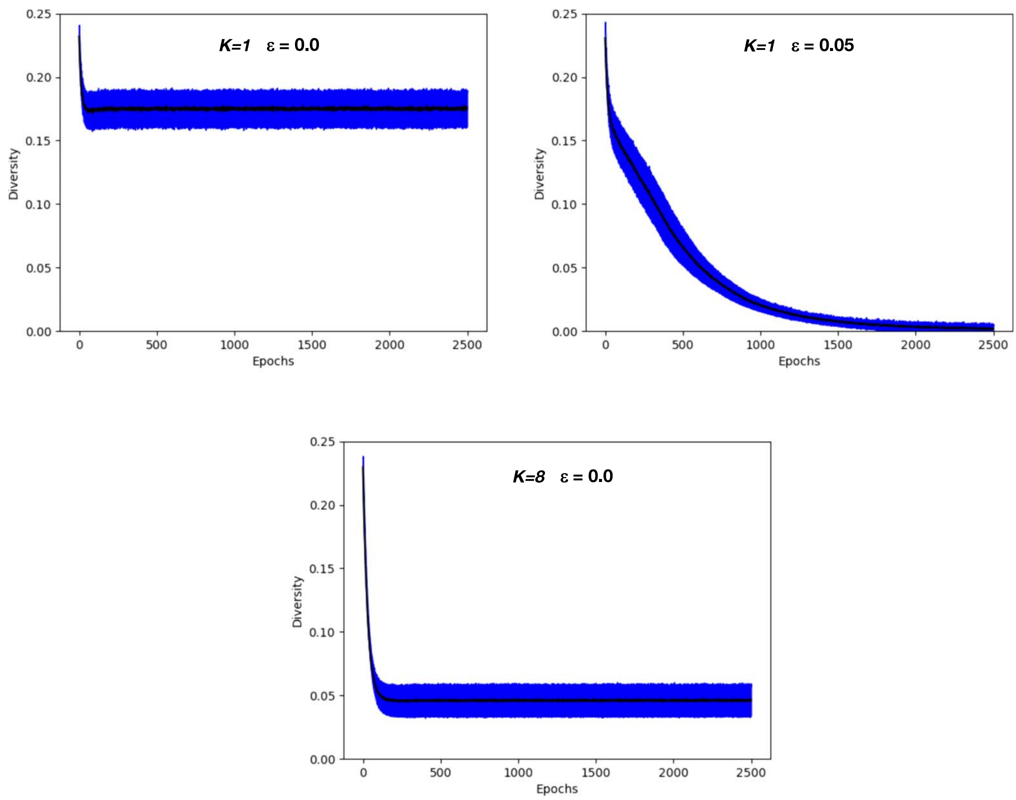

Figure 4. Time series of population diversity (Δ_p) averaged over 25 runs for three different scenarios of the matchmaker-enhanced influence model. All runs employ the parameters given in Table 1, but with $\theta_1 = 36°$ and $\theta_2 = 27°$ for all three, thus ensuring good local (peer-based) conditions for emerging homogeneity. (top left) *K*-nearest-neighbor (KNN) matching with $K = 1$ and $\epsilon = 0$. (bottom) KNN with $K = 8$ and $\epsilon = 0$. (top right) KNN with $K = 1$ and $\epsilon = 0.05$: a 5% chance of random source–influencer pairings. Note the tighter range of *y* axis (diversity) limits compared to Figure 2, to highlight differences between the three matchmaker runs of this figure.

agent B, the chances of either of the two being chosen as the other's nearest neighbor will increase, thus raising the possibility of further attraction between the two and ever-growing similarity. Thus, the matchmaker quickly clusters the agents into self-sustaining islands with little to no external interaction.

As a potential remedy, the matchmaker can find the K nearest neighbors (KNN) in CCV space and then randomly select one of them as an influencer. This reduces the number of islands somewhat, with fewer islands for larger K. A diversity graph of Figure 4 (lower middle) shows the result when K = 8, a much lower final diversity than when K = 1. In these runs, when K = 1, around 15–20 islands form, while only 3–5 arise when K = 8. However, it takes a K of nearly 25 (i.e., half the population size) to insure complete homogeneity.

Alternatively, the addition of a little stochasticity can successfully combat the trend toward diversity. With a probability of ϵ, the matchmaker's chosen source (at any timestep) interacts with a randomly-chosen individual, instead of the best match. Now, the population transitions to homogeneity, with the speed proportional to ϵ. For example, for $\epsilon = 0.01$, it takes 15 to 20 thousand epochs (of 100 timesteps each), whereas increasing to $\epsilon = 0.1$ produces convergence in 1,000 epochs or fewer. Figure 4 (upper right) plots the case for $\epsilon = 0.05$, with full convergence by around 2,000 epochs, on average.

This ϵ-matchmaker model more closely reflects reality in that most people receive recommendations by a combination of (a) nearest-neighbor and other machine-learning algorithms while navigating various online commercial sites, and (b) friends, acquaintances and random contacts both online and offline. And in at least the latter case, where people explicitly interact, the general assumptions of this model should hold: influence can involve assimilation, neutrality or repulsion.

The matchmaker models thus indicate that automated systems whose nudges stem from influencers that are algorithmically-selected for maximum match (and impact), do not, on their own, produce homogeneity. Instead, they form multiple islands of similarity (multiple poles) that resist further mergers. But when coupled with the normal interactions of daily life (and the small attitudinal and behavioral changes that these often entail), the clusters tend to succumb to adhesive forces and eventually merge. These results also agree with many of the studies above, particularly Keijzer and Mäs (2022), showing that recommenders and other forms of personalization can, but do not necessarily, lead to polarization or homogeneity, since early cluster formation can induce positive feedbacks in which the members of numerous small groups only receive influence from one another.

6 The Predictor Agent

Returning to a predominantly local situation (without a matchmaker and its ability to search the entire population for suitable influencers), an alternate model captures the effects of adaptive, goal-oriented systems whose ability to influence depends upon their ability to predict. These *predictive agents* also house a CCV, but one that never changes and thus serves as a static target (i.e., goal) toward which the predictor attempts to push the population.

As in the original model of social influence, at each timestep, a source and influencer agent are stochastically chosen from the population. Only basic agents are chosen as sources, while either basic agents or predictors can be selected as influencers. When choosing an influencer, the *predictor choice probability* (PCP) determines the odds of choosing a predictor. When both source and influencer are basic agents, the dynamics of influence are the same as in the basic peer-to-peer model. However, if the influencer is a predictor, then the strength of its influence depends upon its predictive accuracy.

As summarized in Figure 5, a predictor uses a neural network to map the attributes of another agent (the source) to that agent's expected behaviors. When comparing its prediction of the source's

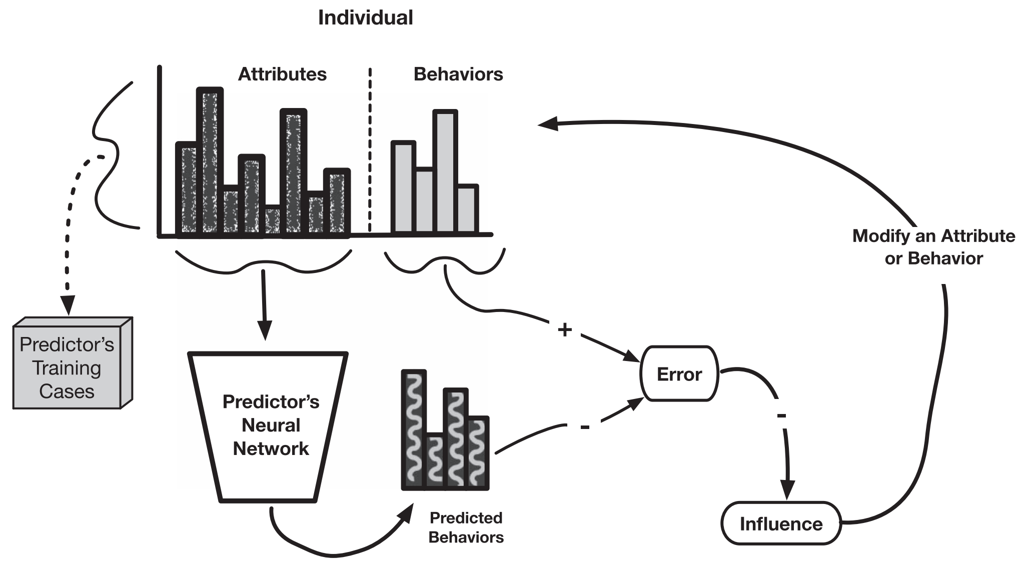

Figure 5. The general behavior of a predictor agent. When predictors encounter basic agents, they cache their attributes and behaviors as inputs and targets, respectively, to comprise training cases (for later use). In addition, they process the attributes of the current agent, producing predicted behaviors. The difference between predicted and actual behaviors yields an error term, which inversely affects the predictor's ability to influence (i.e., modify) a randomly chosen attribute or behavior of the agent in the general direction of the predictor's target composite characteristic vector.

behavior vector to the actual value, the resulting error inversely affects the predictor's influential strength (Γ) according to the following formula:

$$\Gamma = \frac{\omega}{1 + \beta^2 \sqrt{E}} \tag{3}$$

where ω represents the maximum possible strength of a predictor, β is the length of behavior vectors in the CCV, and E is the mean squared error:

$$E = \frac{1}{\beta} \sum_{b \in \beta} (p_b - s_b)^2 \tag{4}$$

where s_b is the bth value of the source's behavior vector, and p_b is bth output of the predictor network.[4] Then, a randomly-chosen characteristic (s_c) of the source's CCV (i.e., either an attribute or a behavior) is modified as follows:

$$s_c = s_c + \Gamma(\rho_c - s_c) \tag{5}$$

where ρ_c is the cth value of the predictor's goal CCV. Note that Equation 5 contains no sign-determining z term (as in Equation 1), thus insuring that the source always changes in the direction of the predictor's CCV.

A predictor agent maintains a buffer of CCVs from the M most recent agents that it has encountered. These constitute the M training cases (see Table 2) for the predictor's neural network, with

4 Because CCVs are unit vectors, E is a fraction, often small. By using $\sqrt{E}$ instead of E, a more significant inverse influence of E on Γ occurs in Equation 3. Multiplying by β^2 further increases that influence, as deemed necessary via experimentation with many alternative relationships between E and Γ.

Table 2. Parameter settings for several runs of the predictor model.

Parameter	Value
Number of attributes (Λ)	10
Number of behaviors (β)	5
Population size (P)	50
Number of predictors (NP)	1
Total timesteps (T)	250,000
Epoch steps (ES)	100
Adaptivity rate (α)	0.25
Maximum predictor strength (ω)	0.25
Similarity trigger proximity (θ_1)	$\{9^\circ, 18^\circ, 27^\circ\}$
Difference trigger proximity (θ_2)	27°
Dimensions of predictor neural network	$[\Lambda, 10, 10, \beta]$
Neural network activation function	sigmoid
Neural network optimizer	stochastic gradient descent (SGD)
Predictor learning rate (λ)	{0.01, 0.1 }
Predictor's memory size (M)	25
Predictor choice probability (PCP)	0.1

the CCV's attributes as inputs and behaviors as target outputs. The timesteps of a run are partitioned into epochs, with ES timesteps per epoch, and each predictor network trains on its M cases at the end of each epoch. Thus, the predictor gradually learns the general correlations between attributes and behaviors in the population. Its ability to predict those relationships then correlates with its ability to nudge the attributes and behaviors of basic agents toward its own goal CCV. This abstractly models any real-world actor (online or offline) whose general ability to **predict** human behavior enhances its powers to **shape** various human characteristics, whether attitudes, preferences or overt actions.

The complete predictor-enhanced algorithm (with parameters from Table 2) is as follows in Steps 1–4:

1. Initialize a population (Φ) of P individuals (NP of which are predictors) to random attribute and behavior vectors.
2. Initialize the weights of each predictor's neural network.
3. Repeat for T timesteps:
   - Select a random source (S) from all basic agents of Φ.

- Select an influencer (I) from Φ, with probability PCP of choosing a predictor.
- If I is a basic agent: {same process as in basic model of social influence}
  - Compare the CCVs of S and I.
  - If S and I are very similar or different, modify one characteristic of S accordingly.
- Else I is a predictor:
  - Add the CCV of S to the training cases of I
  - Run the CCV of S through I's neural network to compute the error (Equation 4)
  - Compute I's influence upon S based on the error (Equation 3)
  - Modify the CCV of S based on that influence (Equation 5)
- If (T modulo ES) = 0: {end of an epoch}
  - Train each predictor's neural network with the M cases in its current training set.
  - Compute and save population diversity

4. Plot the history of population diversity.

Figures 6–9 present the results of four different predictor-model scenarios, each run 25 times. In all runs, θ_2 keeps a constant value of 27°, while θ_1 varies, thus invoking different degrees of

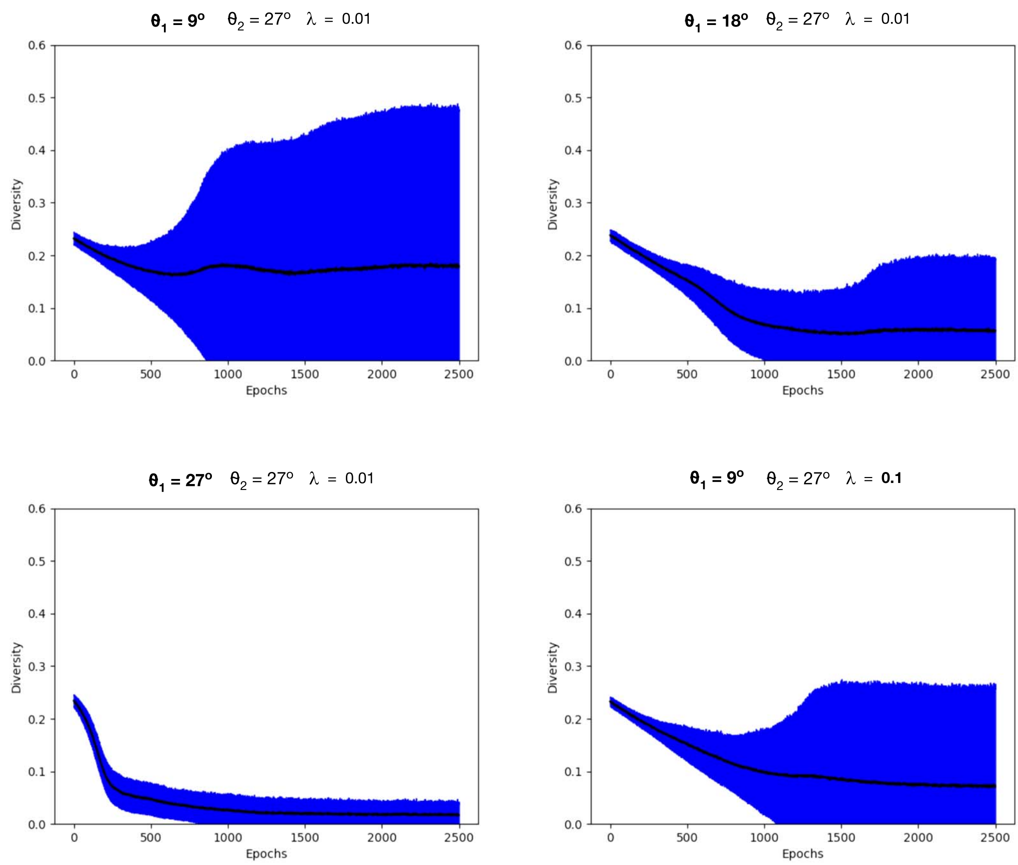

Figure 6. A collection of 25-run averages of diversity for predictor-model runs with $\theta_2 = 27°$, $\theta_1 \in \{9°, 18°, 27°\}$, and $\lambda \in \{0.01, 0.1\}$.

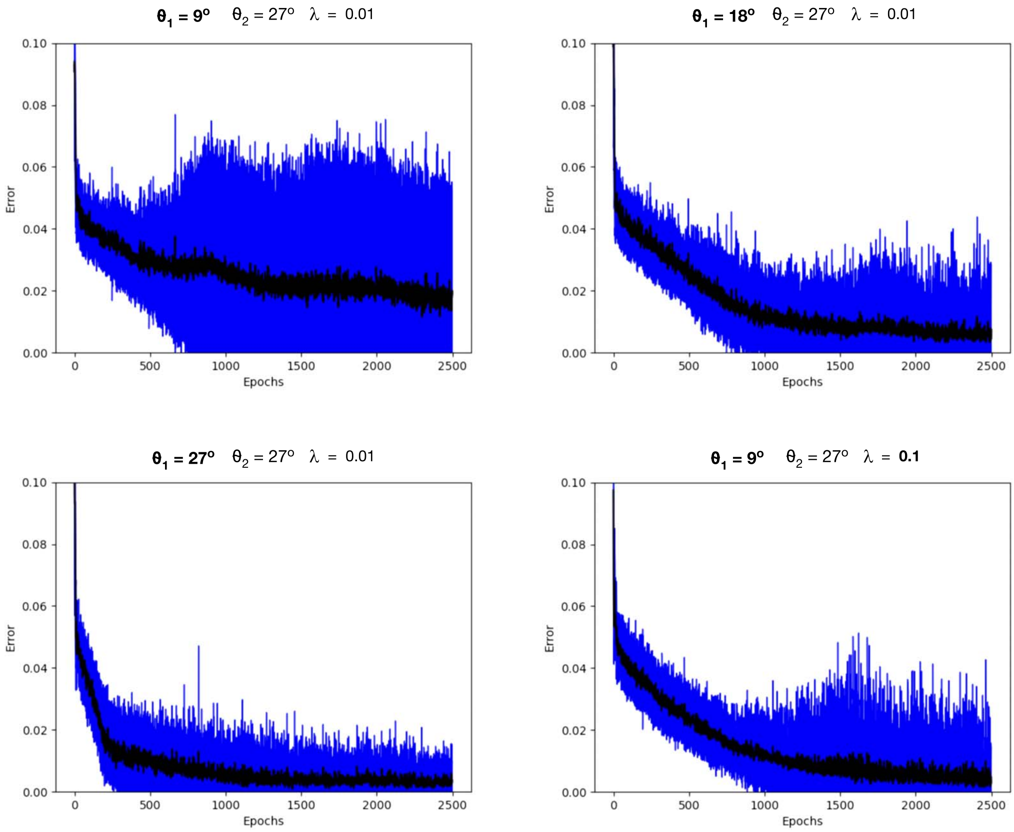

Figure 7. A collection of 25-run averages of prediction error for predictor-model runs with $\theta_2 = 27^\circ$, $\theta_1 \in \{9^\circ, 18^\circ, 27^\circ\}$, and $\lambda \in \{0.01, 0.1\}$.

positive peer influence. In three of the four scenarios, the predictor network's learning rate remains constant, $\lambda = 0.01$, while the final scenario involves a higher value, $\lambda = 0.1$, designed to show that stronger predictor learning can counteract an ominous imbalance between θ_1 (9°) and θ_2 (27°), which clearly favors divergence–runs of the basic model (not shown) with these trigger settings yield a diversity well above 0.8 (78°).

The first scenario ($\theta_1 = 9^\circ, \theta_2 = 27^\circ, \lambda = 0.01$) provides a stark contrast to the corresponding basic-model run. Although, on average, diversity declines slightly from its initial value (Figure 6, upper left), many runs see an increase, though rarely to as high as 0.8. The predictor clearly exerts a significant homogenizing effect as it tries to push all CCVs toward its target value. However, the weak positive peer influence and dominant negative influence confound the predictor's learning problem by maintaining a diverse population with few useful patterns and correlations for the neural network to exploit; the resulting weak learner has limited influence.

In the second scenario, ($\theta_1 = 18^\circ$, $\theta_2 = 27^\circ, \lambda = 0.01$), the increase in θ_1 proves significant, as the population's diversity now shows a clear negative trend (Figure 6, upper right), ending at an average near 0.055 ($\approx 19^\circ$). This seems to support learning, since the predictor's average error drops below 0.01, whereas 0.02 was all it could muster in the first scenario (Figure 7, top row).

One final bump of θ_1, to 27°, produces the third scenario ($\theta_1 = 27^\circ, \theta_2 = 27^\circ, \lambda = 0.01$), in which the combined effects of positive peer influence and the predictor drive diversity down to near 0, with much less variance than in the two previous scenarios (Figure 6, lower left). Note that prediction error (Figure 7, lower left) and diversity drop nearly in tandem, thus hinting of a significant relationship between the two. Compare these runs to those of Figure 2 (upper left),

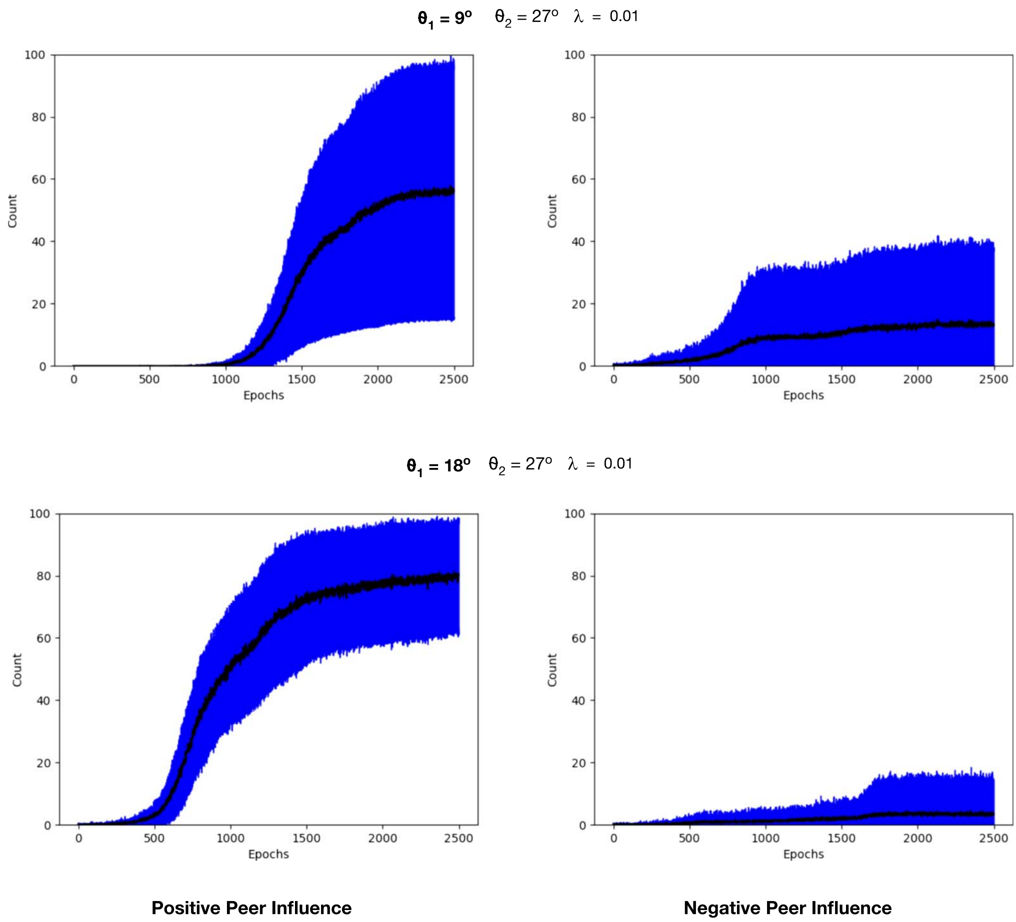

Figure 8. The 25-run averages of positive and negative peer influence for the first two scenarios summarized in the upper rows of Figures 6 and 7.

where $\theta_1 = \theta_2 = 27^\circ$ yields high diversity. The advent of the predictor clearly tips the scales in favor of convergence. Although positive peer influence alone does promote clustering, it still draws CCVs in many different directions and can produce conflicting CCV pairs. Conversely, the predictor pulls them all toward the same target, thus abating most conflict.

Finally, the fourth scenario returns to the trigger settings of the first, but increases the predictor's learning rate by an order of magnitude: $\theta_1 = 9^\circ, \theta_2 = 27^\circ, \lambda = \mathbf{0.1}$. Now, a more aggressive learner manages to restrain negative peer influence, as seen by comparing the upper right plot of Figure 8 with the lower right plot of Figure 9. Thus, when reinforced by the bolder predictor, its positive influence is enough to yield significant convergence. Note, however, that the predictor's error variance elevates considerably (Figure 7, lower right) around the time (1,250 epochs) when negative peer influence makes its most significant jump, to over five updates per epoch (Figure 9, lower right). In this scenario, it seems that the predictor *beats negative influence to the punch* and thereby establishes a sustainable level of conformity.

The plots of peer influence in Figures 8 and 9 illustrate several interesting behaviors. First, as expected, the higher values of θ_1 produce an earlier onset of positive influence, with increases in negative influence occurring soon thereafter. However, with each upgrade of θ_1, the prevalance of negative influence decreases to the point that it barely exceeds a few interactions (per 100 pairings in each epoch) when $\theta_1 = 27^\circ$. The strong role that prediction plays in quelling

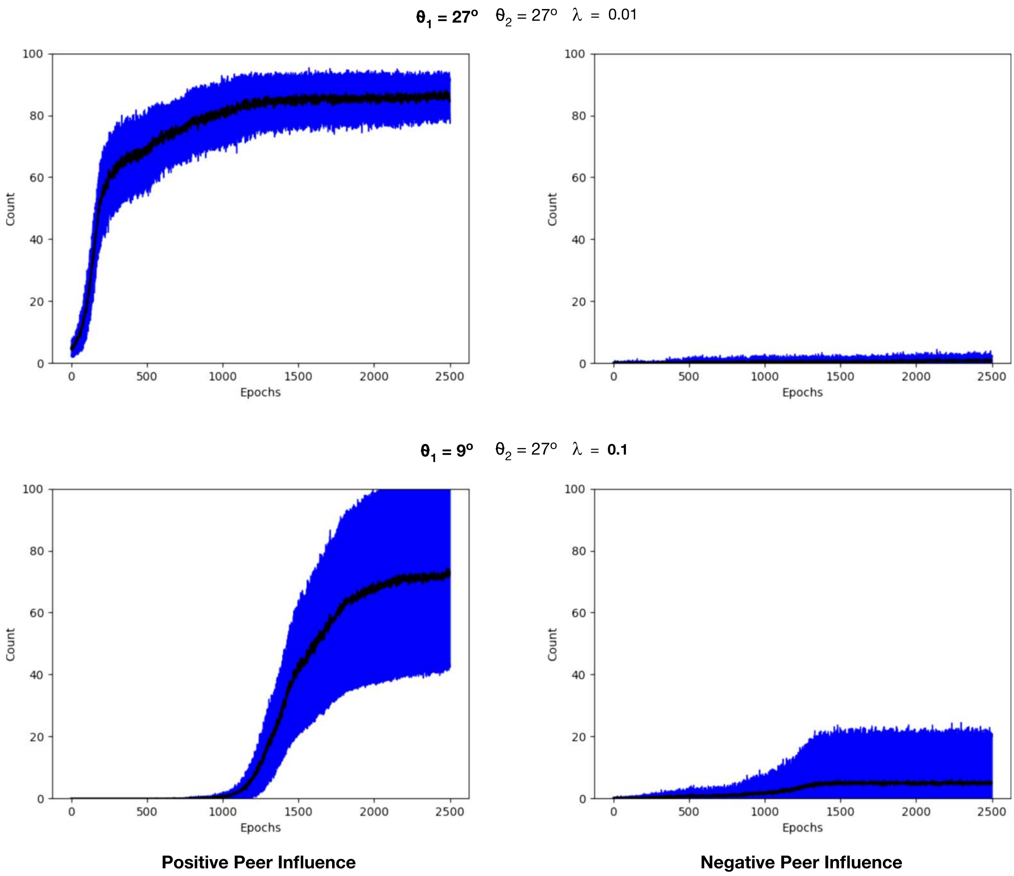

Figure 9. The 25-run averages of positive and negative peer influence for the third and fourth scenarios summarized in the lower rows of Figures 6 and 7.

negative influence becomes obvious in comparing the upper scenario ($\theta_1 = 27°, \theta_2 = 27°, \lambda = 0.01$) of Figure 9 to the pair of plots in Figure 3, which covers basic runs with $\theta_1 = \theta_2 = 27°$, wherein negative influence gains a foothold and drives diversity up above 0.3 ($\approx 46°$) in Figure 2 (upper left).

These results suggest the positive feedback displayed in the causal diagram of Figure 10, wherein declining diversity simplifies the learning problem of the predictor, which can then exert a stronger influence on the population, thereby pushing many CCVs toward the predictor's target CCV, thus reducing diversity even more, which further simplifies the learning problem. Figures 6, 8, and 9 illustrate that a fortification of either positive influence (by increasing θ_1) or prediction (by increasing λ) accelerates the transition to conformity and increases its steady-state level (i.e., decreases the steady-state diversity). This provides good evidence that the causal influences work in both directions, thus producing an abstract example of *the Loop* and its emergence.

7 Conclusion

The combination of these primitive models of multi-directional peer influence and goal-driven, predictor-based nudges provides a very basic abstraction of prevalent forces in today's online world, where individuals experience both high-frequency peer interactions (often with no clear goals but realizing nudges nonetheless), and very directed (though often hidden) shaping by agents whose

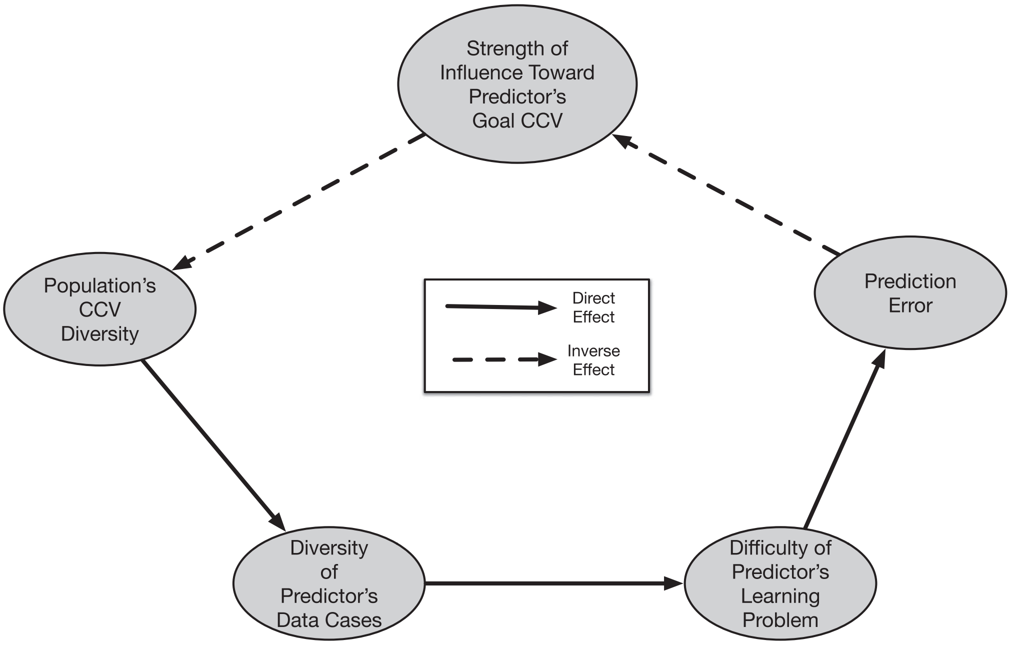

Figure 10. A key positive feedback loop underlying emergent homogeneity. As population diversity declines, the predictor's training cases become more similar and thus easier to learn. The more accurate predictor thus gains greater influence over the population and can prod it more forcefully toward its own CCV goal vector, thus decreasing diversity even more. Note that the presence of an even number of inverse relationships signals a positive feedback.

success depends upon their ability to predict human behavior. As in most agent-based models, the phenomena that emerge depend upon myriad parameter settings. As such, all results can rightly be labeled as *artifacts of the model*.

As Flache et al. (2017) argued, these agent-based models are necessary for getting a better handle on the complex causal relationships and emergent consequences of social influence, but their assumptions should have a solid basis in empirical studies. For example, they discuss how positive influence among interacting individuals has a well-documented history, but repulsive and neutral results are more unclear. The models above incorporate all three, with simple parameters determining the strength of each.

None of the runs above give any deep explanation of, nor prove the inevitability of, the tendency toward conformity and polarization that many see as a dangerous consequence of social media, online commerce and hidden behavior manipulation. Thus, they do not validate the claims of the original motivators of this work: Pariser (2011), Ward (2022) and (tangentially) Zuboff (2019). Rather, they further illustrate the difficulties of drawing strong conclusions about relationships between personalization and diversity reduction. The three models serve mainly to supplement philosophical debates and psychological studies with additional means of computational exploration, with a particular focus on one key aspect that most previous models lack but which Pariser (2011), Ward (2022) and Zuboff (2019) view as serious threats: the ability of online actors to predict our behavior.

By design, these models avoid global control mechanisms in an attempt to show that conformity can emerge from predominantly local interactions. However, each predictor agent does have a goal and can accumulate information about subsets of the population via its random, local pairings with basic agents. Furthermore, the matchmaker introduces significant globality via its ability to search the entire population for the K best matches. But as the results show, these strategic pairings seem to prevent complete convergence while still promoting polarity. Thus, the general effect of real-life matchmakers such as recommender systems and other forms of personalization may not be quite as bad as many anticipate: they may not nudge all agents toward the same island despite their obvious promotion of clusters. Still, the ϵ-matchmaker results show that these clusters may serve as scaffolding for the ascent to pure conformity in the presence of a small degree of random peer-to-peer interaction and influence, as clearly exists in human society. In short, these models indicate that online tools combine with normal human social tendencies to promote conformity, but neither force can do it all alone, except in highly unlikely real-world scenarios where

1. all agents interact randomly with the entire population and have very high susceptibility to positive influence (i.e., high θ_1 in the above models), or
2. all agents interact frequently with the same goal-directed influencer.

As characterized by Ward (2022), *the Loop* relies on this interplay between online nudging and human social tendencies. In the basic peer-to-peer model, the key human predilection is toward associating with other humans and reacting (positively or adversely) to their views. The matchmaker plays the role of an innocuous recommender system, while the predictor represents a goal-directed (often profit-seeking) entity hoping to alter human behavior in very specific ways. Thus, the mapping to Ward's main three entities is not one-to-one. Individuals do map to the basic agents, but aspects of corporations and AI are reflected in both the matchmaker and predictor. The basic results, however, do reflect one of Ward's fundamental concerns: by bowing excessively to influences from myriad online actors or from *people like us* (chosen by us or a mediating AI system), we can lose our individual agency to an extent that online systems can easily predict our behaviors, and thus more easily control us to their own advantage.

The models above can hopefully serve as starting points for more in-depth explorations into filter bubbles, personalization polarization, and interactions between behavior prediction and opinion diversity. One obvious direction involves the introduction of temporality into prediction, such that the goal-directed agent(s) must ascertain and try to exploit the future behaviors of basic agents. Although the concept of prediction need not include time, since it may indicate the ability to deduce some information from other clues, its most interesting connotations involve speculations about the future, and these could surely find a place in an expanded model. Such an enhancement might require a shift of granularity, from the overall characteristics of an agent to its minute-by-minute actions, such as the products that it buys, messages that it posts, etc., as was done in several of the related articles discussed earlier.

Agent-based models often incorporate spatial dimensions to increase realism. Even though the internet has significantly reduced many barriers to interaction, there are still plenty of influences tied to one's geographic location, whether in terms of the individuals one meets in public or the local and regional issues that weigh heavily on daily life. A Californian may care a lot more about water and wildfire issues than an Icelander, despite any transatlantic online links between the two people. Longitude, latitude and nationality still matter. The filter bubble is mainly an online phenomenon, so any addition of space should not completely remove the possibilities for random agent interactions. Rather, it might supplement the model with agent locations that bias at least some of the interactions. It should also associate diverse factors with each region such that an agent's location would bias its characteristics in meaningful ways.

A simple, abstract, and more theoretically satisfying approach to spatial enhancement is the incorporation of graph models (i.e., networks) into these multi-agent simulations, as has been done by many of the projects cited above, going back to French's (1956) seminal model and Axelrod's (1997) classic. Dynamic links based on the evolving inter-agent compatibilities would then provide very tangible evidence of clustering, diversity and homogeneity (e.g., one big clique). Furthermore, if the graph's topology constrains the exposure of the predictor in meaningful ways, this type of model could reveal relationships (a) between an agent's associations and its vulnerability to goal-directed actors, and (b) the dependence of a predictor's performance upon the structure of networks on which it preys. For example, a predictor with interactions restricted to friend links could easily lose global significance if the friends circle their wagons into a tight clique with no connections to dissimilar agents.

Finally, a very intriguing possibility addresses the predictor's learning machinery. Tristan Harris (2021), a former Google researcher, has characterized the online battle for our attention as a *race to the bottom of the brainstem*. Many of the authors cited earlier echo that point (Carr, 2010, 2014; Ward, 2022; Wu, 2016; Zuboff, 2019). This implies that internet actors want to appeal to our primitive (reptilean) instincts, emotions and reactions as much as possible, not to our higher, cognitive faculties (that might generate logical arguments for observing patience, delaying gratification, forgiving, gathering more facts, etc.). These actors often derive immediate benefits from visceral influences and concomitant knee-jerk behaviors, whether in terms of direct purchases or various forms of outrage that often lead to more clicks, screen time and (ultimately) presence in the merchants' virtual showrooms.

The modeling challenge tied to Harris's (2021) insinuation is to implement decision-making systems (e.g., neural networks) that have both reactive and more contemplative options, which are triggered by different aspects of the goal-directed actor's communications. These actors could tailor their messages toward one *brain* region or the other. To make things interesting, tradeoffs (for both basic agents and online actors) would need to exist between impulsive and reasoned behavior. Multiple online entities could then compete for attentional and monetary resources using sophisticated messaging strategies that might devolve into an all-out reptilean appeal but might also transition to a sensible balance that targets both emotion and thought, and has positive consequences in both the short- and longer term. A burgeoning set of potential models exists, and all go well beyond the simple neural networks of the current work.

Whether labeled as filter bubbles, the loop, echo chambers or attitude-consistent information consumption, the problems of emergent polarization and conformity go far beyond the level of *interesting academic pursuits* and *challenging modeling tasks*. These dangerous trends and their geopolitical ramifications threaten humanity in very concrete and terrifying ways, though politicians and technology companies have been slow to act on them. For some of the more notorious world leaders, inaction aids and abets their ongoing schemes, which leverage conformity to more easily control their citizenry. For many tech giants, stockholder earnings drive most decisions, and emerging conformity greases the profit-making machinery. For others, in tech and in politics, an awareness of the actual problem and its scale has not fully gelled. Hopefully, modeling can improve understanding and awareness in the same way that climate models provide essential insights into global warming; because in both cases, the world desperately needs tools for informing proactive choice. We cannot afford to just let things play out.

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